

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY

IN-CLASS MIDTERM, 2:40-3:55 ON 4/1/26

6 numbered problems

Closed-book; no notes, computers, calculators, nor visible phones

Unless explicitly stated otherwise, you may use any result proved in class. If your solution makes use of a theorem from class, make sure to explain why it applies (e.g., if the theorem has hypotheses).

Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

It is not sufficient to state that a problem is identical to one from a previous homework or practice exam, when that is the case.

If you run out of space, use the back side of the paper and indicate this clearly.

Name: _____

Uni: _____

1. (a) Explain what “short exact sequence” means.

- (b) Give an example of a short exact sequence of the form

$$0 \rightarrow \mathbb{Z} \xrightarrow{f} \mathbb{Z} \xrightarrow{g} \mathbb{Z}/3\mathbb{Z} \rightarrow 0.$$

Your answer should define f and g and justify exactness.

- (c) Does there exist a short exact sequence of the form

$$0 \rightarrow \mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0?$$

Justify your answer.

- (d) State the definition of *chain complex* and *chain map*. For a chain complex (A_*, ∂) , state the definition of $H_p(A_*, \partial)$.

2. Let C be a category, and let c , c' , and c'' be objects of C .
- (a) Let $f \in C(c, c')$ a morphism. State the definition of what it means for f to be an isomorphism.

 - (b) Let $f \in C(c, c')$ and $f' \in C(c', c'')$ be isomorphisms. Prove that $f' \circ f$ is also an isomorphism.

 - (c) Let $f, f' \in C(c, c')$ be two isomorphisms, and assume $f \neq f'$. Prove that there exists a $\phi \in C(c, c)$ such that ϕ is an isomorphism and $\phi \neq \text{id}_c$.

3. (a) State the excision theorem.

(b) For a subspace $A \subset X$, let $q : H_p(X) \rightarrow H_p(X, A)$ denote the homomorphism induced by the quotient maps $C_*(X) \rightarrow C_*(X, A)$. Let $i : X \setminus A \rightarrow X$ be the inclusion map, and consider the composition

$$H_p(X \setminus A) \xrightarrow{i_*} H_p(X) \xrightarrow{q} H_p(X, A).$$

Prove that if $A \subset X$ is both open and closed, then this composition is an isomorphism.

(c) With notation as in question (b) above, give an example showing that the composition $q \circ i_*$ need not be an isomorphism in general.

4. (a) State a formula for the homology group $H_p(S^n)$, for all p and all n .

(b) Let $K \subset \mathbb{R}^3$ be a subspace, and assume that K is homeomorphic to S^1 . Determine a formula for $H_p(\mathbb{R}^3, K)$ for all p .

Justify your answer.

5. (a) State the definition of *degree* of a continuous map $f : S^n \rightarrow S^n$.

(b) In class we defined a group homomorphism $\phi : \pi_1(X, x) \rightarrow H_1(X)$ for any based space (X, x) , and stated the theorem that it induces an isomorphism $\text{ab}(\pi_1(X, x)) \rightarrow H_1(X)$ for any path connected X . Explain how the homomorphism $\phi : \pi_1(X, x) \rightarrow H_1(X)$ is defined. *State the definition of the homomorphism. You do not need to prove that it is well defined, nor that it is a group homomorphism.*

(c) Let $f : S^1 \rightarrow S^1$ be a continuous map such that $f(x) = x$ for some x , and assume that $f_*([\lambda]) = [\lambda]^{-1}$ for all $[\lambda] \in \pi_1(S^1, x)$. What can you say about the degree of f ? *Justify your answer.*

6. For each of the following statements, mark if they are true or false.

In this problem, you do not need to justify your answers.

(a) If X is simply connected, then $H_1(X) = 0$.

(b) If $H_1(X) = 0$, then X is simply connected.

(c) If X and Y are homotopy equivalent, then $H_p(X)$ is isomorphic to $H_p(Y)$ for all $p \in \mathbb{Z}$.

(d) If $A \subset X$ and $H_p(X) = 0$, then $H_p(X, A) = 0$.

(e) If $A \subset X$ and $H_p(X) = 0$ and $H_{p-1}(A) = 0$, then $H_p(X, A) = 0$.

(f) If $X \subset \mathbb{R}^4$ and $H_4(\mathbb{R}^4, X) \cong \mathbb{Z}$, then $H_3(X) \cong \mathbb{Z}$.

(g) If $F : \mathbf{Top} \rightarrow \mathbf{Groups}$ is a functor, then $F(\emptyset) \cong \{e\}$.

(h) If $p : X \rightarrow Y$ is a covering map, then $p_* : H_1(X) \rightarrow H_1(Y)$ is injective.